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Carbometalation–Carboxylation of 2,3-Allenols with Carbon Dioxide: A Dramatic Effect of Halide Anion

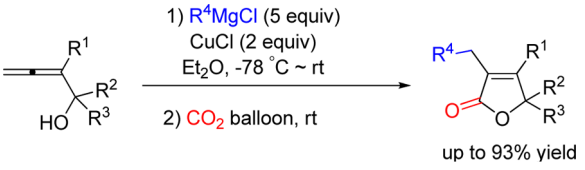
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ABSTRACT



The cyclic organometallic intermediates formed via CuCl-mediated highly regio- and stereoselective carbomagnesiation of 2,3-allenols with Grignard reagents may smoothly react with carbon dioxide to afford 2(5*H*)-furanones. A dramatic effect of the halide anion from the Grignard reagent (Br vs Cl) for CO_2 activation was observed. The reaction proceeded smoothly under mild conditions to afford the products in 58–93% yields.

Much attention has been paid to the development of new and efficient methodologies for the synthesis of 2(5*H*)-furanones,[1] a class of important heterocycles with a broad range of potential biological activities.[2] On the other hand, carbon dioxide is one of the most attractive C1 synthons in organic synthesis as it is a type of renewable, nontoxic, abundant, and economical resource.[3] There are a few reports on the synthesis of 2(5*H*)-furanones from the reaction of alkynols with Grignard reagents with or without Cp_2TiCl_2 followed by quenching with inletting CO_2 gas or pouring into dry ice.[4] Recently, we have developed an efficient regio- and stereospecific CuCl-mediated carbomagnesiation of differently substituted 2,3-allenols with primary, secondary alkyl, or aromatic Grignard reagents followed by iodination to synthesize fully substituted allylic alcohols (Scheme 1).[5] We envisioned that the cyclic intermediates **A** formed in situ in the reaction would react with CO_2 to form γ-hydroxy *Z*-alkenoic carboxylic acid, which would be followed by lactonization to produce butenolides. Herein, we report such a transformation in which a dramatic halide anion effect for CO_2 activation was observed and a CO_2 balloon was used without continuous release of CO_2 to the environment.

After the CuCl-mediated carbometalation of *n*-BuMgBr with allenol **1a**, CO_2 was introduced into reaction vessel by

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(1) For recent reviews, see: (a) Romeo, G.; Iannazzo, D.; Piperno, A.; Romeo, R.; Corsaro, A.; Rescifina, A.; Chiacchio, U. *Mini-Rev. Org. Chem.* **2005**, *2*, 59. (b) Langer, P. *Synlett* **2006**, 3369. (c) Bruckner, R. *Curr. Org. Chem.* **2001**, *5*, 679. (d) Ma, S. *Acc. Chem. Res.* **2003**, *36*, 701. (e) Ma, S. *Pure Appl. Chem.* **2004**, *76*, 651.

(2) (a) Otsuka, H.; Kotani, K.; Bando, M.; Kido, M.; Takeda, Y. *Chem. Pharm. Bull.* **1998**, *46*, 1180. (b) Juan Hikawczuk, V. E.; Roberto Saad, J.; Giordano, O. S.; García, C.; Martín, T.; Martín, V. S.; Sosa, M. E.; Tonn, C. E. *J. Nat. Prod.* **2008**, *71*, 190. (c) Parvatkar, R. R.; D'Souza, C.; Tripathi, A.; Naik, C. G. *Phytochemistry* **2009**, *70*, 128. (d) Pour, M.; Špulák, M.; Buchta, V.; Kubanová, P.; Marie Vopršalová, M.; Wsól, V.; Fáková, H.; Koudelka, P.; Pourová, H.; Schiller, R. *J. Med. Chem.* **2001**, *44*, 2701.

(3) For recent reviews, see: (a) Louie, J. *Curr. Org. Chem.* **2005**, *9*, 605. (b) Sakakura, T.; Choi, J.-C.; Yasuda, H. *Chem. Rev.* **2007**, *107*, 2365. (c) Aresta, M.; Dibenedetto, A. *Dalton Trans.* **2007**, 2975. (d) Correa, A.; Martín, R. *Angew. Chem., Int. Ed.* **2009**, *48*, 6201. (e) Riduan, S. N.; Zhang, Y. *Dalton Trans.* **2010**, *39*, 3347. (f) Huang, K.; Sun, C.; Shi, Z. *Chem. Soc. Rev.* **2011**, *40*, 2435. (g) Martín, R.; Kleij, A. *ChemSusChem.* **2011**, *4*, 1259. (h) Cokoja, M.; Bruckmeier, C.; Rieger, B.; Herrmann, W. A.; Kühn, F. E. *Angew. Chem., Int. Ed.* **2011**, *50*, 8510.

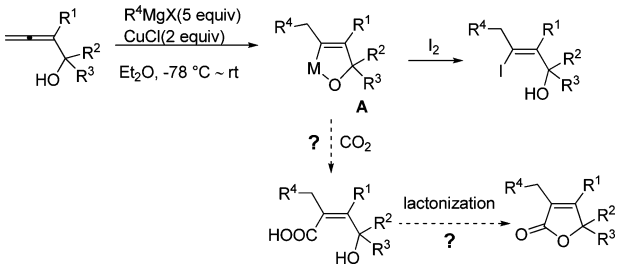
(4) (a) Tan, Z.; Negishi, E.-i. *Org. Lett.* **2006**, *8*, 2783. (b) Ito, T.; Okamoto, S.; Sato, F. *Tetrahedron Lett.* **1990**, *31*, 6399. (c) Tani, K.; Sato, Y.; Okamoto, S.; Sato, F. *Tetrahedron Lett.* **1993**, *34*, 4975. (d) Ishino, Y.; Wakamoto, K.; Hirashima, T. *Chem. Lett.* **1984**, 765. (e) Forgione, P.; Wilson, P. D.; Fallis, A. G. *Tetrahedron Lett.* **2000**, *41*, 17.

(5) Lu, Z.; Ma, S. *Adv. Synth. Catal.* **2007**, *349*, 1225.

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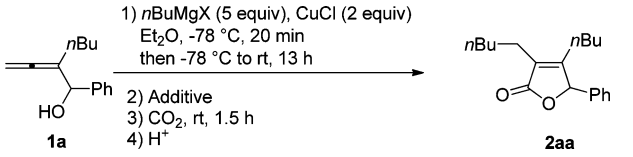
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Scheme 1. Our Previous Work and Proposed Synthesis of 2(5*H*)-Furanones



installation of a CO_2 balloon and **2aa** was formed in 45% yield (Table 1, entry 1). We believed that the reason for the low yield might be the low reactivity of the intermediate **A** toward CO_2. When HMPA or CsF was added as an additive before quenching of the intermediate **A** with CO_2, the yield of **2aa** dropped to 23% (Table 1, entries 2 and 3). To our delight, when LiCl was used, the yield of **2aa** was raised to 79% (Table 1, entry 4). With such an observation of the unique effect of halide anion, we further used *n*-BuMgCl instead of *n*-BuMgBr: interestingly, the yield was further improved to 87% (Table 1, entry 5)!

Table 1. Optimization of Reaction Conditions for the Synthesis of 2(5*H*)-Furanone **2aa**[a]



entry	X	additive (equiv)	NMR yield of **2aa** (%)
1	Br	none	45
2	Br	HMPA (2)	23
3	Br	CsF (2)	23
4	Br	LiCl (2)	79
5[b]	**Cl**	**none**	**85 (87)**[c]

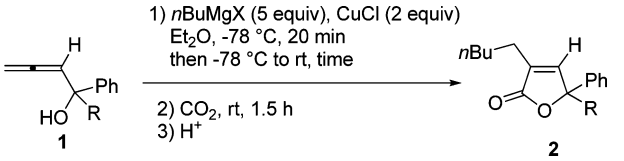
[a] The reaction was conducted using 1 mmol of **1**, 2 mmol of CuCl, 5 mmol of *n*-BuMgBr (2 M in Et_2O), and a CO_2 balloon in 6 mL of Et_2O. [b] The reaction was conducted with 1 mmol of **1**, 2 mmol of CuCl, 5 mmol of *n*-BuMgCl (2 M in Et_2O), and a CO_2 balloon in 6 mL of Et_2O. The first step finished after 15 h. [c] Isolated yield in parentheses.

Subsequently, other organomagnesium chlorides and organomagnesium bromides were compared (Table 2). When *n*-C_4H_9MgBr was used, lactone **2ba** was obtained in low yield (Table 2, entry 1) while with *n*-C_4H_9MgCl, **2ba** was isolated in 72% yield (Table 2, entry 2); the halide anion effect also exists in the reaction of secondary 2,3-allenol **1c** (Table 2, entries 3 and 4).[6]

On the basis of these results, the scope of this reaction was then studied using organomagnesium chlorides.

(6) Metzger, A.; Bernhardt, S.; Manolikakes, G.; Knochel, P. *Angew. Chem., Int. Ed.* **2010**, *49*, 4665.

Table 2. Effect of Halide Anion

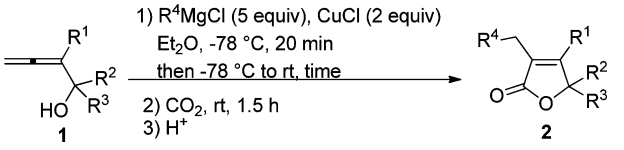


entry	R	*n*-BuMgX	time (h)	yield of **2** (%)
1[a]	CH_3 (**1b**)	$^nC_4H_9MgBr$	13	48[d] (**2ba**)
2[b]	CH_3 (**1b**)	$^nC_4H_9MgCl$	14	72 (**2ba**)
3[a]	H (**1c**)	$^nC_4H_9MgBr$	13	44[d] (**2ca**)
4[b]	H (**1c**)	$^nC_4H_9MgCl$	13	78 (**2ca**)
5[c]	H (**1c**)	$^nC_4H_9MgI$	24	complicated

[a] The reaction was conducted using 1 mmol of **1**, 2 mmol of CuCl, 5 mmol of *n*BuMgBr (2 M in Et_2O), and a CO_2 balloon in 6 mL of Et_2O. [b] The reaction was conducted with 1 mmol of **1**, 2 mmol of CuCl, 5 mmol of BuMgCl (2 M in Et_2O), and a CO_2 balloon in 6 mL of Et_2O. [c] The reaction was conducted using 1 mmol of **1**, 2 mmol of CuCl, 5 mmol of *n*BuMgI (2 M in Et_2O), and a CO_2 balloon in 6 mL of Et_2O. [d] NMR yield.

The results show in Table 3 indicated that 2(5*H*)-furanones were formed in moderate to excellent yields: R^1 can be alkyl, phenyl, benzyl, or H; R^2 may be alkyl, phenyl, methyl, or H. Primary (Table 3, entries 1−3), secondary (Table 3, entries 4−13), and tertiary 2,3-allenols (Table 3, entry 14) all produced the lactones in moderate to good yields. Grignard reagents such as CyMgCl (entry 4), *i*-PrMgCl (entry 5), and PhMgCl (entry 6) were examined in this reaction producing moderate to good yields of 2(5*H*)-furanones. To check the practicality, this reaction was scaled up to 1.12 g (7 mmol)

Table 3. Synthesis of 2(5*H*)-Furanones[a]



entry	$R^1/R^2/R^3$	R^4 MgCl	time (h)	isolated yield (%)
1	nC_6H_{13}/H/H (**1d**)	$^nC_4H_9$	10	75 (**2da**)
2	$^nC_4H_9$/H/H (**1e**)	$^nC_4H_9$	10	73 (**2ea**)
3	Bn/H/H (**1f**)	$^nC_4H_9$	12	88 (**2fa**)
4	H/Ph/H (**1c**)	Cy	13	68 (**2cb**)
5	H/Ph/H (**1c**)	^{i}Pr	13	58 (**2cc**)
6	H/Ph/H (**1c**)	Ph	13	75 (**2cd**)
7	H/4-ClC_6H_4/H (**1g**)	$^nC_4H_9$	12	76 (**2ga**)
8	H/1-naphthyl/H (**1h**)	$^nC_4H_9$	13.5	61 (**2ha**)
9	H/nC_5H_{11}/H (**1i**)	$^nC_4H_9$	12	65 (**2ia**)
10	H/Cy/H (**1j**)	$^nC_4H_9$	14	67 (**2ja**)
11	Ph/Me/H (**1k**)	$^nC_4H_9$	13	76 (**2ka**)
12[b]	Ph/Me/H (**1k**)	$^nC_4H_9$	11	81 (**2ka**)
13	Ph/Ph/H (**1l**)	$^nC_4H_9$	14	82 (**2la**)
14	H/nC_6H_{13}/Me (**1m**)	$^nC_4H_9$	13	64 (**2ma**)

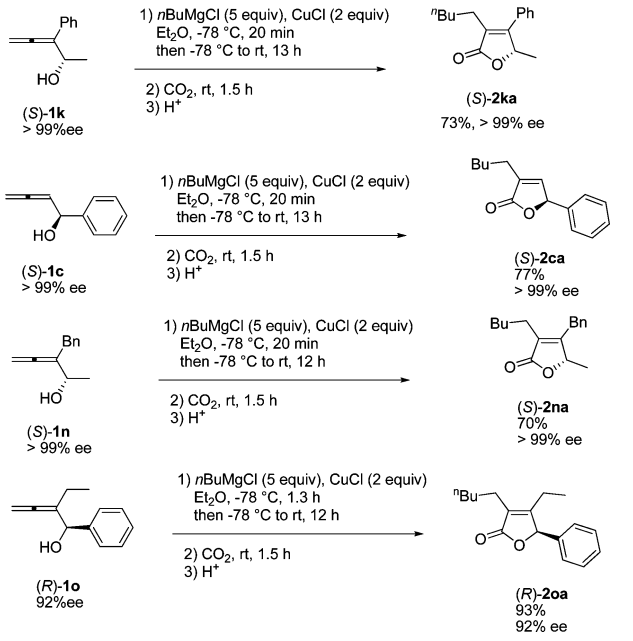
[a] The reaction was conducted with 1 mmol of **1**, 2 mmol of CuCl, 5 mmol of R^4 MgCl (2 M in Et_2O), and a CO_2 balloon in 6 mL of Et_2O. [b] The reaction was conducted using 7 mmol of **1k** (1.12 g scale).

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affording **2ka** in 81% yield (Table 3, entry 12). It should be noted that in some cases the premature protonation product was observed as the minor byproduct and the reaction of **1c** with 3 equiv of *n*-C_4H_9MgCl afforded a lower yield of **2ca** after 24 h.

When the optically active 2,3-allenols *S*-**1k** *S*-**1c**, *S*-**1n4,** and *S*-**1o** were applied, optically active 2(5*H*)-furanones *S*-**2ka**, *S*-**1ca**, *S*-**2na**, and *R*-**2oa** were prepared in 70−92% yields without obvious racemization (Scheme 2).

Scheme 2. Synthesis of Optically Active 2(5*H*)-Furanones

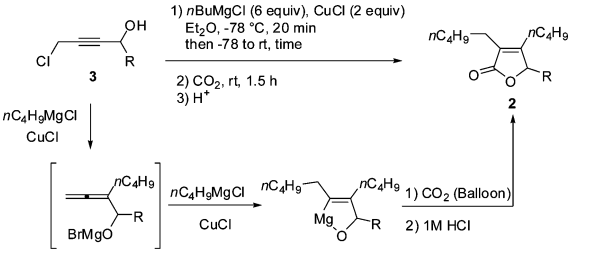


2(5*H*)-Furanones **2ea** and **2aa** may also be synthesized directly from 4-chlorobut-2-yn-1-ol (**3**) with the in situ synthesis of the allenol,[7] carbomagnesiation, carboxylation with CO_2, and lactonization in one pot (Scheme 3).

(7) Li, J.; Kong, W.; Fu, C.; Ma, S. *J. Org. Chem.* **2009**, *74*, 5104.

(8) (a) Xu, D.; Li, Z.; Ma, S. *Chem.—Eur. J.* **2002**, *21*, 5012. (b) Kuang, J.; Ma, S. *J. Org. Chem.* **2009**, *74*, 1763. (c) Kuang, J.; Ma, S. *J. Am. Chem. Soc.* **2010**, *132*, 1786. (d) Kuang, J.; Luo, H.; Ma, S. *Adv. Synth. Catal.* **2012**, *354*, 933. (e) Ye, J.; Li, S.; Chen, B.; Fan, W.; Kuang, J.; Liu, J.; Liu, Y.; Miao, B.; Wan, B.; Wang, Y.; Xie, X.; Yu, Q.; Yuan, W.; Ma, S. *Org. Lett.* **2012**, *14*, 1346.

Scheme 3. Synthesis of 2(5*H*)-Furanones **2ea** and **2aa** in One Step from 4-Chlorobut-2-yn-1-ol



It should be noted that three carbon−carbon bonds and a carbon−oxygen bond were formed in this transformation.

In conclusion, we have developed a very convenient method to synthesize 2(5*H*)-furanones using 2,3-allenols, organomagnesium chlorides, and carbon dioxide provided in the form of a balloon as the starting materials. *It is interesting to observe a dramatic chloride effect of the Grignard reagents for this carboxylation with CO_2*: the yields with RMgCl are much higher than those with RMgBr. Optically active 2(5*H*)-furanones were synthesized from optically active 2,3-allenols. As a result of usefulness of the products and the easy availability of various 2,3-allenols,[5,7,8] the reaction may be potentially useful in organic and medicinal chemistry. Further studies in this area are in progress in our laboratory.

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Supporting Information Available. Spectroscopic data, general procedure, and $^1H/^{13}C$ NMR spectra of all the products. This material is available free of charge via the Internet at http://pubs.acs.org.

The authors declare no competing financial interest.

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